

Overview

Discrete-event simulation in Julia: machine_shop (8 replications of 4000 minutes, seed 20260101)

Title: DiscreteEventSimulation | Model: machine_shop | Seed: 20260101 | Replications: 8 | Horizon: 4000 | Warmup: 400 |
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Overview

A discrete-event simulation study: the engine, the calibration and the experiments, with every number carrying its confidence interval.

MODELS
5

THROUGHPUT
0.133 items_per_hour

CYCLE TIME
MEAN
552.5 minutes

UTILISATION
0.863

COMPLETED
477.5

VALIDATION
validated

REEVALUATION
keep

Design of the study

Seed 20260101
Days 45
Replications 8
Horizon 4000
Warmup 400
Models Any["mmc", "transfer_line", "machine_shop", "inventory", "call_center"]
Objective cycle_time_mean
Online policy offline_first

The models

Catalogue

Model	Title	Entity	Resource	Answers
mmc	ServicePool	customer	server	ServicePoolWithIdenticalServers
transfer_line	TransferLine	part	station	SerialLineWithBuffersAndFailures
machine_shop	JobShop	job	machine	JobShopWithRoutingsReworkScrapAndBreakdowns
inventory	InventoryPosition	order	stock	ReorderPointPolicyWithLeadTimeAndBackorders
call_center	ContactCenter	call	agent	AgentPoolWithPatienceAndPriority

Machine shop over 8 replications

Metric	Mean	Half width	Unit	Rel hw pct
availability	1	0	ratio	0
wait_p95	1034	103.3	minutes	9.987
max_queue_length	44.1	6.49	count	14.7
queue_length_mean	75.5	0.054	count	0.071

Metric	Mean	Half width	Unit	Rel hw pct
cycle_time_mean	552.5	12.7	minutes	2.294
setup	2	0	minutes	0
scrapped	22.9	4.057	count	17.7
service_mean	—	—	minutes	—
throughput	0.133	0.00265	items_per_hour	1.998
events	7762	105.8	count	1.363
span	3599	0.388	count	0.011
reworked	71.2	5.031	count	7.061
wait_mean	552.5	12.7	minutes	2.294
completed	477.5	9.543	count	1.998
wip_mean	4.155	0.061	count	1.476
utilisation	0.863	0.011	ratio	1.285
tardiness_mean	449	11.5	minutes	2.564

DiscreteSim.jl, section overview, seed 20260101